Statistical Learning

Gennaro Basile

October 14, 2025

Contents

1 Introduction to Cluster Analysis **5**
1.1 Clustering versus Classification . . . 5
1.2 Multidimensional Data Representation . . . 5

2 What is Clustering? **5**
2.1 Informal Definition . . . 5
2.2 Key Considerations . . . 6
2.3 Cluster Definition: Cohesion and Separation . . . 6
2.4 The Challenge of Cluster Interpretation . . . 6

3 Graphical Cluster Detection **6**
3.1 Visualization Techniques . . . 7
3.2 Case Study: Air Pollution Data . . . 7

4 Applications of Clustering **8**
4.1 Data Summarization . . . 8
4.2 Customer Segmentation . . . 8
4.3 Social Network Analysis . . . 8
4.4 Preprocessing for Other Data Mining Tasks . . . 8

5 Components of Clustering **8**

6 Proximity Measures **9**
6.1 The Concept of Proximity . . . 9
6.2 Distance Matrix for Numerical Data . . . 9

7 Mathematical Properties of Distance Metrics **9**
7.1 The Formal Definition of a Metric . . . 9
7.2 Relaxed Definitions . . . 10
7.3 Ultrametric Distance . . . 10

8 L_p Norm Distances for Numerical Data **10**
8.1 General L_p Norm . . . 10
8.2 Special Cases . . . 11
8.2.1 L_1: Manhattan (City-Block) Distance . . . 11
8.2.2 L_2: Euclidean Distance . . . 11
8.2.3 L_∞: Chebyshev (Lagrangian) Distance . . . 11
8.2.4 Fractional Norms ($0 < p < 1$) and Zero Norm ($p = 0$) . . . 11
8.3 Geometric Visualization . . . 11
8.4 Generalized L_p Norm with Weights . . . 12
8.5 Normalized Euclidean Distance . . . 12

9 Properties of L_p Norms **13**
9.1 Monotonicity . . . 13
9.2 Translation Invariance . . . 13
9.3 Scale Dependency . . . 13
9.4 Rotation Invariance of Euclidean Distance . . . 13

10 Distance Indices (Relaxed Metrics) **13**

10.1 Squared Euclidean Distance . . . 14
10.2 Relationship to Correlation . . . 14

11 Mahalanobis Distance **14**
11.1 Definition . . . 15
11.2 Interpretation as Normalized Euclidean Distance . . . 15
11.3 Key Properties . . . 15
11.4 Practical Example . . . 15
11.5 Application in PCA-based Clustering . . . 16

12 Similarity Indices for Categorical Data **16**
12.1 Formal Definition of Similarity Index . . . 17
12.2 Relationship to Distance . . . 17
12.3 Application to Numerical Data . . . 17

13 Similarity Indices for Boolean (Binary) Data **17**
13.1 Contingency Table . . . 17
13.2 Common Similarity Coefficients . . . 18
13.2.1 Russell and Rao Index . . . 18
13.2.2 Jaccard Coefficient . . . 18
13.2.3 Sokal-Michener (Simple Matching) Coefficient . . . 18
13.3 Ordering of Coefficients . . . 18
13.4 Should Co-Absences Be Considered? . . . 18
13.5 Other Binary Similarity Measures . . . 19
13.6 Monotonic Relationships . . . 19
13.7 Numerical Example . . . 19

14 Similarity Measures for Multi-Category Categorical Data **20**
14.1 Method 1: One-Hot Encoding . . . 20
14.2 Method 2: Simple Matching on Original Variables . . . 21
14.3 Method 3: Subset-Based Similarity . . . 21
14.4 Numerical Example . . . 21

15 Gower's General Similarity Measure for Mixed Data **22**
15.1 Definition . . . 22
15.2 Variable-Specific Similarities . . . 22
15.2.1 For Binary and Multi-Category Categorical Variables . . . 22
15.2.2 For Continuous Variables . . . 22
15.3 Properties of Gower's Index . . . 22
15.4 Comprehensive Numerical Example . . . 23

16 From Similarity to Distance **23**
16.1 Method 1: Direct Complement . . . 24
16.2 Method 2: Power and Additive Transformations . . . 24
16.3 Practical Consideration . . . 24

17 Final Remarks on Distance Measures **24**
17.1 Measurement Scale of Distances . . . 24
17.2 Objective Evaluation of Distances . . . 25
17.3 The Art of Choosing a Distance Measure . . . 25

18 Summary and Looking Forward **26**
18.1 Students, Classes, and Majors 26
18.1.1 Problem Statement 26
18.1.2 Solution 26
18.1.3 Summary 28

1 Introduction to Cluster Analysis

Cluster Analysis represents one of the fundamental techniques in unsupervised learning, aimed at discovering natural groupings within data without prior knowledge of group membership.

1.1 Clustering versus Classification

The distinction between clustering and classification mirrors the difference between unsupervised and supervised learning paradigms. Consider a child learning to categorize objects: initially, the child observes various items and naturally groups similar ones together based on their characteristics (color, shape, size) without explicit instruction – this is **unsupervised learning** or **clustering**. Later, when a parent explicitly shows the child which category each object belongs to, the learning becomes **supervised**, corresponding to **classification**.
In formal terms:

- **Clustering (Unsupervised):** Given a set of observations, partition them into groups where group membership is unknown a priori
- **Classification (Supervised):** Given labeled training data, learn a function to predict the class of new observations

1.2 Multidimensional Data Representation

Before delving into clustering methods, we must formalize our data structure. A multidimensional dataset $\mathbf{X}$ consists of n records (observations, units, individuals), where each record contains measurements of p features (variables, attributes). Mathematically, for unit i where $i = 1, \ldots, n$, we denote:

$$\mathbf{x}_i' = (x_{i1}, x_{i2}, \ldots, x_{ip})'$$

where $\mathbf{x}_i$ is a p-dimensional column vector representing the i-th observation.
Example: Consider a dataset of stock market information. Each stock (unit) might be characterized by variables such as opening price, closing price, trading volume, market capitalization, and price-to-earnings ratio. If we have $n = 50$ stocks and $p = 5$ features, our data matrix $\mathbf{X}$ would be 50×5.

2 What is Clustering?

2.1 Informal Definition

Clustering encompasses a broad family of methods designed to find subgroups (clusters) within a dataset. The intuitive definition is:

> *Given a set of data points, partition them into groups containing very similar data points that are possibly very dissimilar from those in other groups.*

However, this informal definition immediately raises critical questions:

- What does it mean for points to be "similar" or "dissimilar"?
- How do we measure similarity?
- Which variables should we consider?

2.2 Key Considerations

Scale Dependency: The definition of similarity heavily depends on the measurement scale of features. Consider two individuals characterized by height (in centimeters) and annual income (in euros). Without proper scaling, income differences (ranging in thousands) would dominate height differences (ranging in tens), potentially distorting the clustering results.
All Features Matter: Unless explicitly weighted or selected, all features contribute to the clustering process. This democratic treatment of variables can be both a strength (capturing multidimensional similarity) and a weakness (irrelevant features can obscure true structure).

2.3 Cluster Definition: Cohesion and Separation

Following Gordon (1980), a cluster can be rigorously defined through two complementary properties:

- **Internal Cohesion (Homogeneity):** Points within a cluster should be similar to each other
- **External Isolation (Separation):** Points in different clusters should be dissimilar from each other

Example - Geographical Interpretation: If data points represent geographical locations of houses in a town, clustering might divide the town into compact postal districts. This is an example of *dissection* – partitioning a homogeneous dataset into distinct parts. Each district would have:

- High internal cohesion: houses within a district are geographically close
- High external separation: districts are spatially separated
- Comparable sizes: each contains similar numbers of houses

2.4 The Challenge of Cluster Interpretation

An important caveat, as noted by astronomer Carl Sagan: *"Humans are good at discerning subtle patterns that are really there, but equally so at imagining them when they are altogether absent."* This warning is crucial for cluster analysis:

- The analyst does not know *a priori* the true structure of the data
- Not all clustering solutions correspond to genuine, natural clusters
- Visual patterns can be misleading or artifacts of the chosen method
- Validation and interpretation require domain expertise

3 Graphical Cluster Detection

Before applying formal clustering algorithms, exploratory data analysis through visualization can provide valuable insights into potential cluster structure.

3.1 Visualization Techniques

For low-dimensional data ($p \leq 3$), several graphical methods are effective:

- **Histograms:** Univariate distributions may reveal multimodality, suggesting multiple groups
- **Scatterplots:** Bivariate relationships can show clear groupings in two-dimensional space
- **Density Estimation:** Smooth density estimates help identify regions of high concentration
- **Scatterplot Matrix:** For multivariate data, pairs plots examine all pairwise relationships simultaneously

Example - Body Measurements Data:
Consider a dataset with 20 individuals measured on three variables: Chest, Waist, and Hips (all in inches). A scatterplot matrix enhanced with bivariate density estimates and marginal histograms can reveal:

- Whether measurements cluster into distinct groups (e.g., different body types)
- Correlations between variables
- Presence of outliers

3.2 Case Study: Air Pollution Data

When examining air pollution data across multiple cities with variables such as:

- SOx: Sulfur dioxide content
- TEMP: Temperature
- MANUF: Manufacturing index
- POP: Population
- WIND: Wind speed
- PRECIP: Precipitation
- DAYS: Days with precipitation

A scatterplot matrix might reveal:

1. **Little convincing evidence of group structure:** The data may not naturally separate into distinct clusters
2. **Presence of outliers:** Some cities may have extreme values that could distort clustering
3. **Importance of preprocessing:** Outlier identification and treatment may be crucial before clustering

4 Applications of Clustering

Clustering serves as a foundation for numerous analytical tasks across diverse domains:

4.1 Data Summarization

Reduce large datasets to representative cluster centroids, providing concise insights. For instance, summarizing millions of customer transactions into a few behavioral segments.

4.2 Customer Segmentation

In marketing and e-commerce, clustering identifies groups of customers with similar preferences or behaviors. This enables:

- **Collaborative Filtering:** Recommend products based on what similar customers purchased
- **Targeted Marketing:** Design campaigns tailored to specific segments
- **Product Development:** Understand diverse customer needs

4.3 Social Network Analysis

Detect community structures within networks, identifying groups of highly connected individuals (community detection, group dynamics analysis).

4.4 Preprocessing for Other Data Mining Tasks

Clustering often serves as an intermediate step:

- **Classification:** Build separate models for each cluster
- **Outlier Detection:** Identify points that don't belong to any cluster
- **Feature Engineering:** Use cluster membership as a new feature

5 Components of Clustering

To perform cluster analysis, we need four essential elements:

1. **Data:** The $n \times p$ matrix of observations
2. **Proximity Measure:** A function quantifying similarity or dissimilarity
3. **Clustering Method and Algorithm:** The procedure for forming clusters (hierarchical or non-hierarchical)
4. **Quality Measure:** Criteria for evaluating the final partition

6 Proximity Measures

6.1 The Concept of Proximity

Proximity quantifies how similar (or dissimilar) two objects are. Given two objects O_1 and O_2, we can define either:

- A **similarity** measure: $s_{12} = \text{Sim}(O_1, O_2)$
  - Higher values indicate greater similarity
  - Often normalized to $[0, 1]$
- A **distance** (dissimilarity) measure: $d_{12} = \text{Dist}(O_1, O_2)$
  - Higher values indicate greater dissimilarity
  - Typically non-negative: $d_{12} \geq 0$

These concepts are complementary: high similarity corresponds to low distance, and vice versa.

6.2 Distance Matrix for Numerical Data

Let $\mathbf{X}$ be an $n \times p$ data matrix of numerical variables measured on n units. The distance between units i and j is computed considering their corresponding row vectors:

$$d_{ij} = d(\mathbf{x}'_i, \mathbf{x}'_j)$$

The resulting **distance matrix D** is an $n \times n$ matrix with general element d_{ij} that possesses the following properties:

- **Symmetric:** $d_{ij} = d_{ji}$
- **Non-negative:** $d_{ij} \geq 0$
- **Zero diagonal:** $d_{ii} = 0$
- **Positive semidefinite:** (for certain distance measures)

7 Mathematical Properties of Distance Metrics

7.1 The Formal Definition of a Metric

In geometry, a **distance-metric** between two points corresponding to vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^p$ is a function $d(\mathbf{x}, \mathbf{y})$ satisfying four axioms:

1. **Non-negativity:** $d(\mathbf{x}, \mathbf{y}) \geq 0, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^p$
2. **Identity of indiscernibles:** $d(\mathbf{x}, \mathbf{y}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{y}$
3. **Symmetry:** $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x}), \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^p$
4. **Triangle inequality:** $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^p$

7.2 Relaxed Definitions

By relaxing some axioms, we obtain more general proximity measures:

- **Distance index:** Satisfies non-negativity, identity, and symmetry (but not necessarily triangle inequality)
- **Similarity index:** Typically used for categorical data, satisfying weaker conditions

7.3 Ultrametric Distance

An **ultrametric** is a strengthened form of metric satisfying the *Krassner condition*:

$$d(\mathbf{x}, \mathbf{y}) \leq \max\{d(\mathbf{x}, \mathbf{z}), d(\mathbf{y}, \mathbf{z})\}, \quad \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^p$$

This condition is stronger than the triangle inequality and leads to a remarkable geometric property:
Geometric Interpretation: In ultrametric space, any three points form an isosceles triangle where the base is at most equal to the other two sides (often an equilateral triangle).
Proof of isosceles property: Consider three points with distances $d_1 \geq d_2 \geq d_3$. The Krassner condition implies:

$$\max(d_2, d_3) = d_2 \geq d_1$$

Combined with our ordering assumption $d_1 \geq d_2$, we conclude $d_1 = d_2 \geq d_3$.
Practical Example: Consider 6 consumers and 2 brands (A, B). Define distance as:

$$d_{ij} = \begin{cases} 0 & \text{if consumers } i \text{ and } j \text{ prefer the same brand} \\ 1 & \text{if they prefer different brands} \end{cases}$$

For any three consumers:

- If all prefer the same brand: $d_{12} = d_{13} = d_{23} = 0$ (equilateral)
- If two prefer A and one prefers B: two distances are 0, one is 1 (isosceles with $d_1 = d_2 = 1 > d_3 = 0$)

This distance matrix satisfies the ultrametric property. Ultrametrics naturally arise from hierarchical clustering methods.

8 L_p Norm Distances for Numerical Data

The most widely used family of distance measures for numerical data is the Minkowski or L_p norm family.

8.1 General L_p Norm

The **Minkowski distance** or L_p norm is defined as:

$$d_{ij} = \left[\sum_{k=1}^{p} |x_{ik} - x_{jk}|^p \right]^{1/p}, \quad p \geq 1$$

Different values of p yield different metrics, each with distinct properties and geometrical interpretations.

8.2 Special Cases

8.2.1 L_1: Manhattan (City-Block) Distance

$$d_{ij} = \sum_{k=1}^{p} |x_{ik} - x_{jk}|$$

Interpretation: The distance a taxi would travel in a city with a rectangular grid (hence "city-block"). You can only move along streets (parallel to axes), not diagonally through buildings.
Numerical Example: Consider two points in $\mathbb{R}^2$:

- $\mathbf{x}_1 = (1, 2)$
- $\mathbf{x}_2 = (4, 6)$

$$d_{12}^{(L_1)} = |1 - 4| + |2 - 6| = 3 + 4 = 7$$

8.2.2 L_2: Euclidean Distance

$$d_{ij} = \left[\sum_{k=1}^{p} (x_{ik} - x_{jk})^2\right]^{1/2}$$

Interpretation: The straight-line distance between two points – the most intuitive measure of distance, corresponding to our everyday notion of "how far apart" objects are.
Numerical Example (continued):

$$d_{12}^{(L_2)} = \sqrt{(1 - 4)^2 + (2 - 6)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

8.2.3 L_∞: Chebyshev (Lagrangian) Distance

$$d_{ij} = \max_k |x_{ik} - x_{jk}|$$

Interpretation: The maximum difference across all dimensions. Useful when the largest discrepancy in any single feature is of primary concern.
Numerical Example (continued):

$$d_{12}^{(L_\infty)} = \max\{|1 - 4|, |2 - 6|\} = \max\{3, 4\} = 4$$

8.2.4 Fractional Norms ($0 < p < 1$) and Zero Norm ($p = 0$)

These are technically not metrics (they violate the triangle inequality and are non-convex), but are sometimes used in specialized applications like sparse optimization.

8.3 Geometric Visualization

The "unit sphere" (locus of points at distance r from the origin) varies dramatically with p:

- L_1 **($p = 1$):** Diamond shape (e.g., in 2D: $|x| + |y| = r$)
- L_2 **($p = 2$):** Circle/sphere (e.g., in 2D: $\sqrt{x^2 + y^2} = r$)
- L_∞ **($p = \infty$):** Square/cube (e.g., in 2D: $\max\{|x|, |y|\} = r$)

Example: For a point P at distance r from the origin:

- Under L_1: If $P = (x_P, y_P)$ then $|x_P| + |y_P| = r$. When $x_P = y_P$, we get $2|x_P| = r$.
- Under L_∞: We have $\max\{|x_P|, |y_P|\} = r$, forming a square.

8.4 Generalized L_p Norm with Weights

The **weighted Minkowski distance** incorporates variable-specific weights:

$$d_{ij} = \left[\sum_{k=1}^{p} w_k |x_{ik} - x_{jk}|^p\right]^{1/p}, \quad p \geq 1, \quad w_k \geq 0, \quad \sum_k w_k = 1$$

Purpose: Weights allow the analyst to emphasize certain variables over others based on domain knowledge or importance.
Example: In customer segmentation, you might assign:

- Higher weight ($w = 0.4$) to purchase frequency
- Moderate weight ($w = 0.3$) to average transaction value
- Lower weight ($w = 0.3$) to time since last purchase

As Anderberg (1973) noted: *"The choice of weights is not a matter for neat mechanical devices; it is one of the principal means through which the analyst can shape the analysis to his objectives."*

8.5 Normalized Euclidean Distance

To handle variables with different scales, the **normalized Euclidean distance** standardizes by variable-specific spread:

$$d_{ij} = \left[\sum_{k=1}^{p} \frac{1}{\sigma_k^2}(x_{ik} - x_{jk})^2\right]^{1/2}$$

where σ_k^2 is the variance of the k-th variable.
Effect: Variables with large variance don't disproportionately dominate the distance calculation.
Numerical Example:
Consider two variables:

- Variable 1 (height in cm): $\sigma_1 = 10$, $x_{i1} = 170$, $x_{j1} = 180$
- Variable 2 (weight in kg): $\sigma_2 = 15$, $x_{i2} = 70$, $x_{j2} = 85$

Standard Euclidean: $d = \sqrt{(170-180)^2 + (70-85)^2} = \sqrt{100 + 225} = \sqrt{325} \approx 18.03$
Normalized Euclidean: $d = \sqrt{\frac{(170-180)^2}{10^2} + \frac{(70-85)^2}{15^2}} = \sqrt{\frac{100}{100} + \frac{225}{225}} = \sqrt{1+1} = \sqrt{2} \approx 1.41$
Notice how normalization prevents the weight variable (with larger raw differences) from dominating.

9 Properties of L_p Norms

9.1 Monotonicity

The L_p norm is a **non-decreasing function of** p (by Jensen's inequality):

$$L_1 \geq L_2 \geq \cdots \geq L_\infty$$

Implication: Manhattan distance always yields values at least as large as Euclidean distance for the same points.

9.2 Translation Invariance

The L_p norm is **invariant to variable translation** (shifting all values of a variable by a constant):
If $\mathbf{x}_i^* = \mathbf{x}_i + \mathbf{c}$ and $\mathbf{x}_j^* = \mathbf{x}_j + \mathbf{c}$, then $d(\mathbf{x}_i^*, \mathbf{x}_j^*) = d(\mathbf{x}_i, \mathbf{x}_j)$.
Practical consequence: Centering variables (subtracting means) does not affect distances.

9.3 Scale Dependency

The L_p norm is **NOT invariant to linear transformations** (scaling):
If we multiply a variable by a constant, distances change.
Critical implication: Variables measured on different scales (e.g., income in thousands of euros vs. age in years) will contribute differently to the distance. This often necessitates standardization before clustering.

9.4 Rotation Invariance of Euclidean Distance

The **Euclidean metric is invariant to orthogonal transformations** (rotations and reflections):
If $\mathbf{Q}$ is an orthogonal matrix ($\mathbf{Q}'\mathbf{Q} = \mathbf{I}$), then:

$$d(\mathbf{Q}\mathbf{x}_i, \mathbf{Q}\mathbf{x}_j) = d(\mathbf{x}_i, \mathbf{x}_j)$$

Proof:

$$\begin{aligned} d^2(\mathbf{Q}\mathbf{x}_i, \mathbf{Q}\mathbf{x}_j) &= (\mathbf{Q}\mathbf{x}_i - \mathbf{Q}\mathbf{x}_j)'(\mathbf{Q}\mathbf{x}_i - \mathbf{Q}\mathbf{x}_j) \\ &= (\mathbf{x}_i - \mathbf{x}_j)'\mathbf{Q}'\mathbf{Q}(\mathbf{x}_i - \mathbf{x}_j) \\ &= (\mathbf{x}_i - \mathbf{x}_j)'(\mathbf{x}_i - \mathbf{x}_j) \\ &= d^2(\mathbf{x}_i, \mathbf{x}_j) \end{aligned}$$

Practical significance: If we rotate our coordinate system (e.g., via PCA), Euclidean distances remain unchanged.

10 Distance Indices (Relaxed Metrics)

A **distance index** satisfies only the first three metric axioms (non-negativity, identity, symmetry) but may violate the triangle inequality.

10.1 Squared Euclidean Distance

The square of the Euclidean distance is a distance index:

$$d_{ij}^2 = \sum_{k=1}^{p} (x_{ik} - x_{jk})^2$$

Why it's useful despite not being a metric:

- **Additive property:** For partitions of variables $\{1, \ldots, p'\}$ and $\{p'+1, \ldots, p\}$:

$$\sum_{k=1}^{p'} (x_{ik} - x_{jk})^2 + \sum_{k=p'+1}^{p} (x_{ik} - x_{jk})^2 = \sum_{k=1}^{p} (x_{ik} - x_{jk})^2$$

- **Computational efficiency:** Avoids square root calculations
- **Theoretical properties:** Direct connection to variance decomposition in k-means clustering

10.2 Relationship to Correlation

For **standardized vectors** $\mathbf{z}_x$ and $\mathbf{z}_y$ (mean 0, variance 1), the squared Euclidean distance relates to the linear correlation coefficient ρ_{xy}:

$$d_{xy}^2 = 2p(1 - \rho_{xy})$$

Derivation:

$$\begin{aligned}
d_{xy}^2 &= \sum_{k=1}^{p} (z_{xk} - z_{yk})^2 \\
&= \sum_{k=1}^{p} (z_{xk}^2 - 2z_{xk}z_{yk} + z_{yk}^2) \\
&= \sum_{k=1}^{p} z_{xk}^2 + \sum_{k=1}^{p} z_{yk}^2 - 2\sum_{k=1}^{p} z_{xk}z_{yk} \\
&= p + p - 2p\rho_{xy} = 2p(1 - \rho_{xy})
\end{aligned}$$

Interpretation:

- When $\rho_{xy} = 1$ (perfect positive correlation): $d_{xy}^2 = 0$
- When $\rho_{xy} = 0$ (no correlation): $d_{xy}^2 = 2p$
- When $\rho_{xy} = -1$ (perfect negative correlation): $d_{xy}^2 = 4p$

11 Mahalanobis Distance

The **Mahalanobis distance** incorporates the covariance structure of the data, accounting for correlations between variables and their different scales simultaneously.

11.1 Definition

For two vectors $\mathbf{x}_i, \mathbf{x}_j \in \mathbb{R}^p$:

$$d_{ij} = c\left[(\mathbf{x}_i - \mathbf{x}_j)'\mathbf{S}^{-1}(\mathbf{x}_i - \mathbf{x}_j)\right]^{1/2}$$

where:

- $\mathbf{S}$ is the $p \times p$ sample covariance matrix
- $c = \sqrt{\frac{n}{n-1}}$ is a corrective factor (often omitted in practice)

11.2 Interpretation as Normalized Euclidean Distance

The Mahalanobis distance can be viewed as a Euclidean distance in a transformed space where:

- Variables are standardized to have unit variance
- Correlations between variables are removed

Matrix decomposition perspective: If $\mathbf{S} = \mathbf{L}\mathbf{L}'$ (Cholesky decomposition), then:

$$d_{ij} = \|\mathbf{L}^{-1}(\mathbf{x}_i - \mathbf{x}_j)\|_2$$

This shows that Mahalanobis distance is the Euclidean distance after linear transformation $\mathbf{L}^{-1}$.

11.3 Key Properties

1. **Invariant to non-singular linear transformations:** If $\mathbf{y}_i = \mathbf{A}\mathbf{x}_i + \mathbf{b}$ where $\mathbf{A}$ is non-singular, the Mahalanobis distance in the transformed space equals that in the original space (after updating the covariance matrix accordingly).

2. **Reduces to normalized Euclidean when variables are uncorrelated:** If $\mathbf{S}$ is diagonal (i.e., $\mathbf{S} = \text{diag}(\sigma_1^2, \ldots, \sigma_p^2)$), then:

$$d_{ij} = \left[\sum_{k=1}^{p} \frac{(x_{ik} - x_{jk})^2}{\sigma_k^2}\right]^{1/2}$$

   which is exactly the normalized Euclidean distance.

3. **Scale-invariant:** Automatically handles variables on different scales

11.4 Practical Example

Consider measuring similarity between students based on:

- X_1: Math score (0-100)
- X_2: Verbal score (0-100)

Suppose the covariance matrix is:

$$\mathbf{S} = \begin{pmatrix} 100 & 60 \\ 60 & 100 \end{pmatrix}$$

The correlation is $\rho = 0.6$ (moderately positive).
Two students:

- Student A: $(x_{A1}, x_{A2}) = (70, 60)$
- Student B: $(x_{B1}, x_{B2}) = (80, 65)$

Euclidean distance:

$$d_{AB}^{(\text{Euclidean})} = \sqrt{(70-80)^2 + (60-65)^2} = \sqrt{100+25} = \sqrt{125} \approx 11.18$$

Mahalanobis distance:
First, compute $\mathbf{S}^{-1}$:

$$\mathbf{S}^{-1} = \frac{1}{100 \cdot 100 - 60 \cdot 60} \begin{pmatrix} 100 & -60 \\ -60 & 100 \end{pmatrix} = \frac{1}{6400} \begin{pmatrix} 100 & -60 \\ -60 & 100 \end{pmatrix}$$

Then:

$$\begin{aligned} d_{AB}^{(\text{Mahalanobis})} &= \sqrt{(10,5) \cdot \frac{1}{6400} \begin{pmatrix} 100 & -60 \\ -60 & 100 \end{pmatrix} \begin{pmatrix} 10 \\ 5 \end{pmatrix}} \\ &= \sqrt{\frac{1}{6400}(10,5) \begin{pmatrix} 700 \\ -100 \end{pmatrix}} \\ &= \sqrt{\frac{7000-500}{6400}} = \sqrt{\frac{6500}{6400}} \approx 1.008 \end{aligned}$$

Notice that the Mahalanobis distance is much smaller than the Euclidean distance because it accounts for the positive correlation between Math and Verbal scores (students with high Math scores tend to have high Verbal scores, so the difference is "expected").

11.5 Application in PCA-based Clustering

A common workflow:

1. Perform PCA on the original data to obtain uncorrelated principal components
2. Compute Euclidean distances on the PC scores
3. This gives equal weight to all PCs, unlike using PCs weighted by their eigenvalues
4. The result approximates Mahalanobis distance in the original space

12 Similarity Indices for Categorical Data

When data are categorical rather than numerical, we cannot directly compute distances based on arithmetic differences. Instead, we use **similarity indices**.

12.1 Formal Definition of Similarity Index

A **similarity index** for a finite set of objects $u_i \in U$ is a function $S(u_i, u_j) = S_{ij}$ mapping $U \times U$ to $\mathbb{R}^1$ and satisfying:

1. **Non-negativity:** $S_{ij} \geq 0, \quad \forall u_i, u_j \in U$
2. **Normalization:** $S_{ii} = 1, \quad \forall u_i \in U$
3. **Symmetry:** $S_{ij} = S_{ji}, \quad \forall u_i, u_j \in U$

Note the weaker identity property: $\mathbf{x} = \mathbf{y} \Rightarrow S(\mathbf{x}, \mathbf{y}) = 0$ (not an if-and-only-if).

12.2 Relationship to Distance

The **dissimilarity index** is simply the complement:

$$\delta_{ij} = 1 - S_{ij}$$

Important distinction: Similarity indices are typically applied to *objects* (represented by categorical features) rather than to vectors of numerical measurements.

12.3 Application to Numerical Data

Similarity indices can also be applied to numerical data by treating each variable as categorical (matched/not matched):
Set $s_{ijk} = 1$ if $x_{ik} = x_{jk}$ and $s_{ijk} = 0$ otherwise, then average:

$$S_{ij} = \frac{1}{p} \sum_{k=1}^{p} s_{ijk}$$

This simple matching approach discards magnitude information but can be robust to outliers.

13 Similarity Indices for Boolean (Binary) Data

For binary variables (presence/absence, yes/no, 1/0), we construct a 2×2 contingency table summarizing agreements and disagreements.

13.1 Contingency Table

For two objects u_i and u_j measured on p binary variables:

	u_j		
u_i	1	0	Total
1	a	b	$a+b$
0	c	d	$c+d$
Total	$a+c$	$b+d$	p

where:

- a: number of variables where both have value 1 (co-presences)
- b: number of variables where $u_i = 1, u_j = 0$
- c: number of variables where $u_i = 0, u_j = 1$
- d: number of variables where both have value 0 (co-absences)

13.2 Common Similarity Coefficients

13.2.1 Russell and Rao Index

$$S_{ij}^{(RR)} = \frac{a}{p}$$

Interpretation: Proportion of variables where both objects have value 1.
Characteristic: Penalizes co-absences and individual presences equally.

13.2.2 Jaccard Coefficient

$$S_{ij}^{(\text{Jaccard})} = \frac{a}{a+b+c}$$

Interpretation: Proportion of variables with at least one presence where both have presences.
Characteristic: Ignores co-absences (d), focusing only on shared presences.
Widely used in: Ecology (species presence/absence), text mining (document similarity), recommendation systems.

13.2.3 Sokal-Michener (Simple Matching) Coefficient

$$S_{ij}^{(SM)} = \frac{a+d}{p}$$

Interpretation: Proportion of variables where objects agree (either both present or both absent).
Characteristic: Treats presences and absences symmetrically.

13.3 Ordering of Coefficients

It can be shown that:

$$S_{ij}^{(RR)} \leq S_{ij}^{(\text{Jaccard})} \leq S_{ij}^{(SM)}$$

Proof sketch:

- $S^{(RR)} \leq S^{(\text{Jaccard})}$ because $a \leq a$ and $p \geq a+b+c$
- $S^{(\text{Jaccard})} \leq S^{(SM)}$ because ignoring d (co-absences) can only decrease similarity when $d > 0$

13.4 Should Co-Absences Be Considered?

The choice between coefficients depends on whether **co-absences contain useful information**:
Arguments for including co-absences (use Simple Matching):

- When absence is as informative as presence
- Example: Medical symptoms (not having certain symptoms is diagnostically relevant)

Arguments for excluding co-absences (use Jaccard):

- When absences are uninformative or numerous
- Example: Document similarity (two documents not containing the word "aardvark" doesn't make them similar)
- Example: Species ecology (two sites not having a rare species is uninformative)

13.5 Other Binary Similarity Measures

S3: Rogers and Tanimoto (1960)

$$S_{ij} = \frac{a+d}{a+2(b+c)+d}$$

Weights mismatches more heavily than Simple Matching.
S4: Sneath and Sokal (1973)

$$S_{ij} = \frac{a}{a+2(b+c)}$$

Ignores co-absences and double-weights mismatches.
S5: Gower and Legendre (1986)

$$S_{ij} = \frac{a+d}{\sqrt{[a+\frac{1}{2}(b+c)+d][a+\frac{1}{2}(b+c)+d]}}$$

Geometric mean-based weighting.
S6: Gower and Legendre (1986) - alternative

$$S_{ij} = \frac{a}{\sqrt{[a+\frac{1}{2}(b+c)][a+\frac{1}{2}(b+c)]}}$$

13.6 Monotonic Relationships

Some coefficients are **monotonically related**, meaning they produce the same ordering of object pairs:

- S2 (Jaccard), S4, and S6 are monotonically related
- S1 (Simple Matching), S3, and S5 are monotonically related

Implication: Within each group, the choice affects numerical values but not the clustering structure (for methods based solely on distance ordering).

13.7 Numerical Example

Consider three individuals measured on 10 binary variables:

Individual	V_1	V_2	V_3	V_4	V_5	V_6	V_7	V_8	V_9	V_{10}
A	1	1	0	1	0	0	1	1	0	1
B	1	1	0	0	0	1	1	0	0	1
C	0	0	1	0	1	1	0	0	1	0

For A and B:

- $a = 4$ (variables 1,2,7,10 both have 1)
- $b = 2$ (variables 4,8 A has 1, B has 0)
- $c = 1$ (variable 6 A has 0, B has 1)
- $d = 3$ (variables 3,5,9 both have 0)

- $p = 10$

Coefficients:

$$S_{AB}^{(RR)} = \frac{4}{10} = 0.40$$
$$S_{AB}^{(\text{Jaccard})} = \frac{4}{4+2+1} = \frac{4}{7} \approx 0.571$$
$$S_{AB}^{(SM)} = \frac{4+3}{10} = 0.70$$

For A and C:

- $a = 0$ (no variables where both have 1)
- $b = 6$ (variables 1,2,4,7,8,10)
- $c = 4$ (variables 3,5,6,9)
- $d = 0$ (no variables where both have 0)

Coefficients:

$$S_{AC}^{(RR)} = \frac{0}{10} = 0.00$$
$$S_{AC}^{(\text{Jaccard})} = \frac{0}{0+6+4} = 0.00$$
$$S_{AC}^{(SM)} = \frac{0+0}{10} = 0.00$$

Interpretation: A and C are completely dissimilar (never agree on any variable), while A and B share several features.

14 Similarity Measures for Multi-Category Categorical Data

When categorical variables have more than two categories, we need different approaches.

14.1 Method 1: One-Hot Encoding

Transform each m-category variable into m binary indicator variables:
Example: "Color" with categories {Red, Green, Blue} becomes three binary variables:

- Is_Red: 1 if Red, 0 otherwise
- Is_Green: 1 if Green, 0 otherwise
- Is_Blue: 1 if Blue, 0 otherwise

Then apply binary similarity measures (e.g., Jaccard).
Considerations:

- **Dimension expansion:** If you have q categorical variables with $m_1, \ldots, m_q$ categories, the dimension increases from q to $\sum_{j=1}^{q} m_j$
- **Weighting issues:** Variables with more categories get more "votes" unless corrected
- **Which similarity measure?** Jaccard is often preferred (co-absences are uninformative: not being Red and not being Red doesn't indicate similarity)

14.2 Method 2: Simple Matching on Original Variables

Allocate a score s_{ijk} of 0 or 1 for each variable k based on whether individuals i and j have the same category:

$$s_{ijk} = \begin{cases} 1 & \text{if } x_{ik} = x_{jk} \\ 0 & \text{otherwise} \end{cases}$$

Then compute the average:

$$S_{ij} = \frac{1}{p} \sum_{k=1}^{p} s_{ijk}$$

Advantage: Simple and doesn't inflate dimensionality.
Disadvantage: Treats all mismatches equally (no partial credit for "close" categories).

14.3 Method 3: Subset-Based Similarity

Divide all possible outcomes of each variable into **mutually exclusive subsets of categories**, then check if the two categories are in the same subset.
Application: Linguistic Analysis
This approach has been successfully applied to study language relationships. Linguists identify core-meaning word categories (e.g., numbers, body parts, kinship terms) that are universal across cultures. They then group words into subsets based on phonetic or etymological similarity.
Example: Consider the word for "mother" across languages:

- Subset 1: {madre (Italian), mère (French), madre (Spanish), mãe (Portuguese)} - Romance languages
- Subset 2: {mother (English), Mutter (German), moeder (Dutch)} - Germanic languages
- Subset 3: {mat' (Russian), matka (Polish)} - Slavic languages

Two languages get $s_{ijk} = 1$ for this variable if their words belong to the same subset, 0 otherwise.

14.4 Numerical Example

Three individuals described by two categorical variables:

Individual	Education	Occupation
A	Bachelor	Engineer
B	Bachelor	Teacher
C	Master	Engineer

Method 2 (Simple Matching):

$$S_{AB} = \frac{1}{2}(1+0) = 0.5 \quad \text{(agree on Education only)}$$
$$S_{AC} = \frac{1}{2}(0+1) = 0.5 \quad \text{(agree on Occupation only)}$$
$$S_{BC} = \frac{1}{2}(0+0) = 0.0 \quad \text{(agree on nothing)}$$

15 Gower's General Similarity Measure for Mixed Data

When a dataset contains a mixture of numerical and categorical variables, Gower's (1971) coefficient provides a unified framework.

15.1 Definition

$$S_{ij} = \frac{\sum_{k=1}^{p} w_{ijk} s_{ijk}}{\sum_{k=1}^{p} w_{ijk}}$$

where:

- $w_{ijk} \in \{0, 1\}$ is a validity indicator:
  - $w_{ijk} = 1$ if the comparison is valid
  - $w_{ijk} = 0$ if data is missing or comparison is invalid (e.g., co-absence in binary data when we want to exclude it)
- s_{ijk} is the similarity for variable k based on its measurement scale

15.2 Variable-Specific Similarities

15.2.1 For Binary and Multi-Category Categorical Variables

$$s_{ijk} = \begin{cases} 1 & \text{if } x_{ik} = x_{jk} \\ 0 & \text{otherwise} \end{cases}$$

15.2.2 For Continuous Variables

$$s_{ijk} = 1 - \frac{|x_{ik} - x_{jk}|}{R_k}$$

where R_k is the range of variable k: $R_k = \max_i x_{ik} - \min_i x_{ik}$.
Interpretation: This is the complement of the city-block distance after scaling each variable to unit range $[0, 1]$.

15.3 Properties of Gower's Index

1. $S_{ij} = 1$ if and only if individuals i and j have identical values for all numerical variables and identical categories for all categorical variables
2. $S_{ij} = 0$ if there is complete disagreement on all variables
3. $S_{ij} \in [0, 1]$ for all pairs
4. For datasets with only binary variables (excluding co-absences), Gower's index reduces to the Jaccard coefficient
5. For datasets with only numerical variables, Gower's index is equivalent to the complement of the average normalized city-block distance:

$$S_{ij} = 1 - \frac{1}{p} \sum_{k=1}^{p} \frac{|x_{ik} - x_{jk}|}{R_k}$$

15.4 Comprehensive Numerical Example

Consider three customers characterized by mixed data:

Customer	Age (years)	Income (k euro)	Gender	Loyalty
A	25	35	M	Yes
B	30	40	M	No
C	45	80	F	Yes

Variable ranges:

- Age: $R_{\text{Age}} = 45 - 25 = 20$
- Income: $R_{\text{Income}} = 80 - 35 = 45$

Computing S_{AB}:

1. Age: $s_{AB,\text{Age}} = 1 - \frac{|25-30|}{20} = 1 - \frac{5}{20} = 0.75$
2. Income: $s_{AB,\text{Income}} = 1 - \frac{|35-40|}{45} = 1 - \frac{5}{45} \approx 0.889$
3. Gender: $s_{AB,\text{Gender}} = 1$ (both M)
4. Loyalty: $s_{AB,\text{Loyalty}} = 0$ (Yes vs No)

Assuming all comparisons are valid ($w_{ijk} = 1$ for all k):

$$S_{AB} = \frac{0.75 + 0.889 + 1 + 0}{4} = \frac{2.639}{4} \approx 0.660$$

Computing S_{AC}:

1. Age: $s_{AC,\text{Age}} = 1 - \frac{|25-45|}{20} = 1 - 1 = 0$
2. Income: $s_{AC,\text{Income}} = 1 - \frac{|35-80|}{45} = 1 - 1 = 0$
3. Gender: $s_{AC,\text{Gender}} = 0$ (M vs F)
4. Loyalty: $s_{AC,\text{Loyalty}} = 1$ (both Yes)

$$S_{AC} = \frac{0 + 0 + 0 + 1}{4} = 0.25$$

Interpretation: Customers A and B are more similar (0.660) than A and C (0.25), despite A and C sharing loyalty status, because A and C differ greatly in age and income.

16 From Similarity to Distance

Often, clustering algorithms require dissimilarities or distances rather than similarities. Several transformation strategies exist.

16.1 Method 1: Direct Complement

If $\mathbf{S}$ is a similarity matrix with elements $S_{ij} \in [0, 1]$ and $\mathbf{S}$ is non-negative definite, then:

$$d_{ij} = \sqrt{1 - S_{ij}}$$

defines a **Euclidean distance**.
Proof sketch: Non-negative definiteness of $\mathbf{S}$ ensures that $1 - S_{ij}$ can be embedded in Euclidean space.
Note: Most similarity coefficients (S1-S5 in our earlier table, except S6) have positive semidefinite similarity matrices (Gower and Legendre, 1986), so this transformation is valid.

16.2 Method 2: Power and Additive Transformations

For any dissimilarity matrix $\mathbf{\Delta}$ with elements δ_{ij}, the following transformations can yield Euclidean distances:

1. **Power transformation:** $d_{ij} = \sqrt{\delta_{ij}^2 + c_1}$ for appropriate constant c_1
2. **Additive transformation:** $d_{ij} = \delta_{ij} + c_2$ for appropriate constant c_2

These transformations can "correct" non-Euclidean dissimilarities.

16.3 Practical Consideration

The choice of transformation can affect clustering results. When possible:

- Check if your dissimilarity matrix is Euclidean (via eigenvalue analysis)
- Use transformations that preserve ranking if your clustering method is rank-based
- Be aware that squared transformations emphasize large dissimilarities

17 Final Remarks on Distance Measures

17.1 Measurement Scale of Distances

Ordinal Scale:

- Most distance measures preserve ordering of similarities
- The actual numerical values are less important than their relative magnitudes
- Different distance functions may produce different rankings

Ratio Scale:

- A true zero exists (distance zero means identity)
- However, even when variables are measured on ratio scales, the computed distances are not necessarily on a ratio scale
- Example: Euclidean distance on ratio-scaled variables has meaningful zero and ratios, but this doesn't automatically transfer through all transformations

17.2 Objective Evaluation of Distances

For three vectors $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^p$, we can objectively state that "$\mathbf{x}$ is closer to $\mathbf{y}$ than to $\mathbf{z}$" if:

$$|x_k - y_k| \leq |x_k - z_k|, \quad \text{for } k = 1, 2, \ldots, p$$

with at least one strict inequality.
Example:

- $\mathbf{x} = (2, 5, 8)$
- $\mathbf{y} = (3, 6, 9)$
- $\mathbf{z} = (5, 7, 12)$

Component-wise comparisons:

$$\begin{aligned} |x_1 - y_1| &= 1 < |x_1 - z_1| = 3 \\ |x_2 - y_2| &= 1 < |x_2 - z_2| = 2 \\ |x_3 - y_3| &= 1 < |x_3 - z_3| = 4 \end{aligned}$$

$\mathbf{x}$ is objectively closer to $\mathbf{y}$ than to $\mathbf{z}$ under any L_p norm.

17.3 The Art of Choosing a Distance Measure

There is no universally "best" distance measure. The choice depends on:

1. **Data type:** Numerical vs. categorical vs. mixed
2. **Scale:** Are all variables on comparable scales?
3. **Distribution:** Gaussian vs. skewed vs. heavy-tailed
4. **Correlation structure:** Are variables independent or correlated?
5. **Domain knowledge:** What makes objects "similar" in your application?
6. **Outliers:** Robustness to extreme values (Manhattan more robust than Euclidean)
7. **Interpretability:** Can you explain the distance measure to stakeholders?

General recommendations:

- **Euclidean:** Default for continuous data when variables are standardized and approximately independent
- **Manhattan:** More robust to outliers; suitable when variables have different scales
- **Mahalanobis:** When variables are correlated; accounts for covariance structure
- **Jaccard:** Binary or categorical data where co-absences are uninformative
- **Gower:** Mixed data types (numerical + categorical)

18 Summary and Looking Forward

In this first part of cluster analysis, we have established the foundational concepts:

- **Clustering** is the unsupervised task of partitioning data into homogeneous groups
- **Proximity measures** (distances for numerical data, similarities for categorical data) quantify how alike objects are
- Various **distance functions** (L_p norms, Mahalanobis) suit different data characteristics
- **Similarity indices** (Jaccard, Simple Matching, Gower) handle categorical and mixed data
- The choice of proximity measure profoundly impacts clustering results

With these building blocks in place, we are now prepared to explore **hierarchical clustering methods**, which build nested partitions by iteratively merging or splitting clusters based on proximity. These methods will leverage the distance and similarity measures we have studied to construct dendrograms representing the hierarchical structure of data.

18.1 Students, Classes, and Majors

18.1.1 Problem Statement

We consider m students, n classes, and p majors. Each student can be in any number of classes (typically 3 to 6) and can have any number of majors (commonly 0, 1, or 2).
The data about students' classes and majors are represented by two matrices:

- An $m \times n$ matrix C, where

$$C_{ij} = \begin{cases} 1 & \text{if student } i \text{ is in class } j \\ 0 & \text{if student } i \text{ is not in class } j \end{cases}$$

- An $m \times p$ matrix M, where

$$M_{ij} = \begin{cases} 1 & \text{if student } i \text{ is in major } j \\ 0 & \text{if student } i \text{ is not in major } j \end{cases}$$

Part (a): Let E be the n-vector with E_i being the enrollment in class i. Express E using matrix notation, in terms of the matrices C and M.
Part (b): Define the $n \times p$ matrix S where S_{ij} is the total number of students in class i with major j. Express S using matrix notation, in terms of the matrices C and M.

18.1.2 Solution

Part (a): Finding the enrollment vector E
We need to find the enrollment in each class. The enrollment in class j is simply the total number of students enrolled in that class.

Looking at matrix C, each column j represents a particular class. The entry $C_{ij} = 1$ if student i is in class j, and $C_{ij} = 0$ otherwise. Therefore, to count how many students are in class j, we need to sum all the entries in column j:

$$E_j = \sum_{i=1}^{m} C_{ij}$$

In matrix notation, this is equivalent to multiplying C by a vector of all ones. Specifically, if we let $\mathbf{1}_m$ denote the m-dimensional vector of all ones, then:

$$E = C^T \mathbf{1}_m$$

Verification of dimensions:

- C is $m \times n$, so C^T is $n \times m$
- $\mathbf{1}_m$ is $m \times 1$
- Therefore, $C^T \mathbf{1}_m$ is $n \times 1$, which matches the dimension of E

Understanding the operation: When we compute $C^T \mathbf{1}_m$, the j-th entry of the result is the dot product of the j-th row of C^T (which is the j-th column of C) with $\mathbf{1}_m$. This dot product sums all entries in the j-th column of C, giving us exactly the number of students in class j.

Part (b): Finding the matrix S

We need to find S_{ij}, which represents the number of students who are both in class i and in major j.

A student k contributes to S_{ij} if and only if:

- Student k is in class i: $C_{ki} = 1$
- Student k is in major j: $M_{kj} = 1$

Therefore, we need to count how many students k satisfy both conditions:

$$S_{ij} = \sum_{k=1}^{m} C_{ki} \cdot M_{kj}$$

This is precisely the definition of matrix multiplication! Notice that:

- Column i of C contains 1's for students in class i
- Column j of M contains 1's for students in major j
- The dot product of these two columns counts students in both

In matrix notation:

$$S = C^T M$$

Verification of dimensions:

- C is $m \times n$, so C^T is $n \times m$
- M is $m \times p$
- Therefore, $C^T M$ is $n \times p$, which matches the required dimension of S

Understanding the operation: The (i, j) entry of $C^T M$ is computed as:

$$(C^T M)_{ij} = \sum_{k=1}^{m} (C^T)_{ik} \cdot M_{kj} = \sum_{k=1}^{m} C_{ki} \cdot M_{kj}$$

This sum counts exactly those students k for whom both $C_{ki} = 1$ (student k is in class i) and $M_{kj} = 1$ (student k is in major j), which is precisely what we want for S_{ij}.

18.1.3 Summary

- **Part (a):** $E = C^T \mathbf{1}_m$, where $\mathbf{1}_m$ is the vector of all ones
- **Part (b):** $S = C^T M$

Both solutions exploit the binary nature of the matrices: multiplying binary indicators and summing gives us counts of entities satisfying multiple conditions simultaneously.